Codility

Screen Report: Anonymous

Test Name:

Check out Codility training tasks

Summary Timeline

Tasks summary

Task	Effective time spent	Score
AbsDistinct Python	20 min	100%

Total score

100%

Assessment integrity

No issues

See less

Integrity safeguards

Signal	Issues 0	Action
Network IP check The candidate used one network during the assessment	Not found	

Tasks Details

Easy

1. **AbsDistinct**
Compute number of distinct absolute values of sorted array elements.

Task Score	Correctness	Performance
100%	100%	100%

Task description

A non-empty array A consisting of N numbers is given. The array is sorted in non-decreasing order. The *absolute distinct count* of this

Solution

Programming language used: Python

Time spent on task: 20 minutes

array is the number of distinct absolute values among the elements of the array.

For example, consider array A such that:

```
A[0] = -5
A[1] = -3
A[2] = -1
A[3] =  0
A[4] =  3
A[5] =  6
```

The absolute distinct count of this array is 5, because there are 5 distinct absolute values among the elements of this array, namely 0, 1, 3, 5 and 6.

Write a function:

```
def solution(A)
```

that, given a non-empty array A consisting of N numbers, returns absolute distinct count of array A.

For example, given array A such that:

```
A[0] = -5
A[1] = -3
A[2] = -1
A[3] =  0
A[4] =  3
A[5] =  6
```

the function should return 5, as explained above.

Write an **efficient** algorithm for the following assumptions:

- N is an integer within the range [1..100,000];
- each element of array A is an integer within the range [−2,147,483,648..2,147,483,647];
- array A is sorted in non-decreasing order.

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Notes: *not defined yet*

Task timeline Preview

Code: 09:42:34 UTC, py, final, score: **100** show code in pop-up

```python
# you can write to stdout for debugging purposes, e.g.
# print("this is a debug message")

def solution(A):
    # Implement your solution here
    """
    abs_array =[abs(x) for x in A ]
    distinct_values = len(set(abs_array))
    return distinct_values
    """

    #optimizing it
    left=0
    right=len(A)-1
    count = 0
    visited = None

    while left<=right:
        left_abs = abs(A[left])
        right_abs=abs(A[right])

        if left_abs > right_abs:
            current=left_abs
            left+=1
        elif left_abs<right_abs:
            current=right_abs
            right-=1
        else:
            current=left_abs
            left+=1
            right-=1

        if current!=visited:
            count+=1
            visited=current
    return count
```

09:22:48 09:42:35

Away from Codility tab **0%**

Analysis summary

The solution obtained perfect score.

Analysis

Detected time complexity: **O(N) or O(N*log(N))**

expand all **Example tests**

▶ example example test	✓ OK

expand all **Correctness tests**

▶ one_element	✓ OK
▶ two_elements	✓ OK
▶ same_elements	✓ OK
▶ simple	✓ OK
▶ simple_no_zero	✓ OK
▶ simple_no_same	✓ OK
▶ simple_no_negative	✓ OK
▶ simple_no_positive	✓ OK
▶ arith_overlow	✓ OK
▶ medium_chaotic1	✓ OK
▶ medium_chaotic2	✓ OK

expand all **Performance tests**

▶ long_sequence_no_negative	✓ OK
▶ long_sequence_no_positive	✓ OK
▶ long_sequence	✓ OK